



Pressure-induced novel phases with the high- T_c superconductivity in zirconium dihydride

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ABSTRACT

The unprecedented structures in hydrides could be uncovered under high pressure, which probably possess unusual properties, such as high- T_c superconductivity. Here, the structural transition of ZrH_2 at 0–500 GPa are comprehensively explored by using the evolutionary algorithm in the universal structure predictor combined with first-principles calculations. Three new phases are found: $Pnma$ - ZrH_2 , $P-3m1$ - ZrH_2 and $R-3m$ - ZrH_2 , with dynamic stability. Moreover, the sequence of phases of ZrH_2 under pressure is $I4/mmm \rightarrow P4/nmm \rightarrow Pnma \rightarrow P-3m1 \rightarrow R-3m$ with the zero-point energy effects. Structural feature, electronic properties, bonding characters and superconductivity of all the ZrH_2 are analyzed in detail. More importantly, the superconducting critical temperature T_c values for $P-3m1$ at 440 GPa and $R-3m$ at 500 GPa are 17.70–20.14 K and 26.68–30.03 K, respectively, which stems from the strong electron-phonon coupling.

1. Introduction

Ashcroft proposed that if metallic hydrogen is realized under high pressure, it should have high superconducting transition temperature (T_c) [1–3]. However, metallic hydrogen is not observed experimentally until 360 GPa [4–6]. At lower pressure, one approach to discovering new high-temperature superconductors is to apply chemical pre-compression [7]. H_3S is one typical hydride to exhibit high-temperature superconductivity at high pressures. It is discovered through computational predictions and subsequent experimental verification [8,9]. At about 155 GPa, H_3S is found to exhibit a T_c of approximately 203 K, and the high- T_c superconductivity in H_3S is believed to arise from the strong coupling between the sulfur and hydrogen sublattices leading to dramatic modifications in their electronic and vibrational properties [9]. Another high- T_c high-pressure hydride is LaH_{10} ($T_c = 257$ – 274 K at 250 GPa) with a sodalite-like structure, where the La atoms sit in a face-centered cubic (fcc) lattice, the H atoms occupy the voids or cages, and specific arrangement and bonding of hydrogen within the structure play a crucial role in determining the physical and chemical properties

[10]. In addition, theoretical and experimental results show that YH_6 and YH_9 are high- T_c high pressure hydrides [11–13]. For hydrides, high pressure not only can produce the extraordinary compositions which are different from common stoichiometries under atmospheric pressure, but also could yield the new configurations with unexpected properties for the ambient hydrides.

High pressure can reduce the distance between atoms, which leads to the novel phase transition of hydrides. Because of the great difference in electronegativity, the early transition metals have a tendency to readily react with hydrogen to form hydrides under ambient pressure, such as ScH_2/ScH_3 [14,15], YH_2/YH_3 [16], AcH_2 [17], VH_2 [18], NbH_2 [18], TiH_2 [19], ZrH_2 [20] and HfH_2 [21,22]. ScH_2 is found to transform from $Fm-3m$ into $C2/m$ phase under the pressure of 65 GPa and $Fm-3m-ScH_2$ possesses the T_c of 38.11 K at 30 GPa [14,15]. The study of AcH_2 shows that some electrons resided at the H atom transfer back to the Ac atom under pressure, thereby driving the structural transition [16]. Subsequently, VH_2 and NbH_2 undergo the structural transition from the $Fm-3m$ (cubic) structure to the $Pnma$ (orthorhombic) structure around 50 GPa with the T_c of several Kelvins [18]. With regard to group IVB

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element dihydrides, TiH_2 and HfH_2 are not superconductive at ambient pressure. However, TiH_2 and HfH_2 undergo phase transitions upon compression and T_c increases compared with the atmospheric pressure [21–24]. The T_c values of Cmma-TiH_2 at 250 GPa and P-3m1-HfH_2 at 500 GPa can reach 6 K and 19.74 K, respectively. Zr is the element between Ti and Hf in the periodic table, which stimulate the researchers to probe into the structural transition and the superconductivity of zirconium dihydride under high pressure.

ZrH_2 exhibits a distortion from cube structure to tetragonal structure under atmospheric pressure, which has been studied in detail and determined by predecessors [25]. The tetragonal distortion can have significant effects on their electronic properties, including the density of states (DOS) near the Fermi level, which is mainly due to the splitting of double degenerate bands in $\Gamma - \text{L}$ direction and the movement of Van Hove singularity (VHS) above Fermi level [26]. The Jahn-Teller effect, driven by the interactions between electrons and the crystal lattice, can explain the distortion from cubic to tetragonal structure in ZrH_2 [27–30]. Subsequently, Huang *et al.* found that ZrH_2 undergoes a structural transformation from the $I4/mmm$ phase to the $P4/nmm$ phase at 103 GPa [20]. However, Li *et al.* proposed that the phase transition for ZrH_2 did not occur up to 150 GPa [31]. This promoted us to clarify the controversy on the phase transition of ZrH_2 . In addition, it remain unclear whether the novel phases of ZrH_2 with excellent superconductivity yield under higher pressure (>150 GPa). These two aspects inspire us to uncover the structural transition of ZrH_2 within a broad pressure range, in quest to explore its novel structures and superconductivity.

In the present work, we comprehensively studied the structural transition of ZrH_2 under the pressure range of 0–500 GPa using the evolutionary algorithm of Universal Structure Predictor (USPEX) [32–34] combined with first-principles calculations [35,36]. The structural transition sequence of ZrH_2 is $I4/mmm \rightarrow P4/nmm \rightarrow Pnma \rightarrow P-3m1 \rightarrow R-3m$ with the zero-point energy effects and the transition pressures are 117, 212, 231 and 444 GPa, respectively. Under high pressure, besides two previously known phases ($I4/mmm$ and $P4/nmm$), three new structures ($Pnma$, $P-3m1$ and $R-3m$) are firstly uncovered in ZrH_2 . Moreover, the structural feature, electronic properties, bonding characters and superconductivity for five stable ZrH_2 phases were comprehensively investigated. Notably, the superconducting critical temperature T_c values for $P-3m1$ at 440 GPa and $R-3m$ at 500 GPa are 17.70–20.14 K and 26.68–30.03 K, respectively, which can be attributed to the strong electron-phonon coupling. The present work provides a perfect test bed to deepen the understanding of the pressure-induced phase and superconductivity in ZrH_2 under high pressure.

2. Computational methods

To search for the energetically stable and meta-stable crystalline structures for ZrH_2 under high pressure, an unbiased structure prediction with simulation cells containing 1–8 formula units at 0–500 GPa are performed by using evolutionary algorithm as implemented in the USPEX code [32–34]. Thirty structures were randomly generated in the first generation and new structures were generated in the next generation by variational operator inheritance (50 %), soft mutation (20 %), and lattice mutation (30 %).

All the enthalpies and electronic properties were calculated within the Vienna Ab initio Simulation Package (VASP) [37]. The electron-ion interaction was described using the projector augmented wave (PAW) [38] pseudopotentials and the valence electrons of Zr and H were treated as $4s^2 4p^6 4d^2 5s^2$ and $1s^1$. Additionally, the exchange-correlation function is based on the Perdew–Burke–Ernzerhof (PBE) functional [39] within the generalized gradient approximation (GGA) [40]. A plane-wave cutoff energy of 800 eV and k-point meshes of $2\pi \times 0.03 \text{ \AA}^{-1}$ are chosen to obtain the required energy convergence ($< 1 \text{ meV/atom}$). The phonon dispersion curve, the phonon density and the electron-phonon coupling were calculated to investigate the superconductivity using Quantum-Espresso package [41,42]. The wave-function

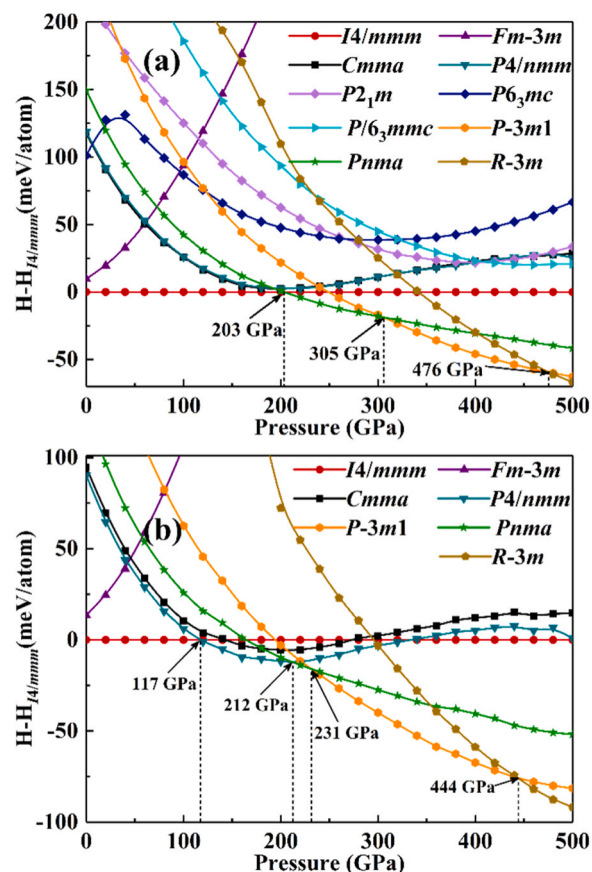


Fig. 1. Enthalpy-pressure relation of ZrH_2 with respect to $I4/mmm$ phase (a) without zero-point energy effects and (b) with zero-point energy effects.

kinetic energy cutoffs of 80 Ry and charge density kinetic energy cutoff of 320 Ry in combination with the pseudopotentials derived from of the SSSP precision library [43]. Moreover, the adopted k and q -point meshes were summarized in Table S1 of the Supplementary Information.

3. Results and discussion

3.1. Structural Transitions

To obtain all the low-energy configurations of ZrH_2 over the pressure range of 0–500 GPa, we have performed the structure predictions with simulation sizes ranging from one to eight ZrH_2 formula units (f.u.). The subsequent analysis of the low-energy crystalline configurations confirmed eight candidate structures, which are denoted as $Fm-3m$, $I4/mmm$, $P4/nmm$, $Cmma$, $Pnma$, $P2_1/m$, $P-3m1$ and $R-3m$. In addition, the ground-state structures $P6_3/mmc$ and $P6_3mc$ for other transition metal dihydrides has been also considered [44]. The experimentally fluorite structure ($Fm-3m$) and tetragonal structure ($I4/mmm$) at ambient pressure are successfully reproduced, which justifies the validity of the USPEX methodology adopted in the structural searches of ZrH_2 . The candidate crystal structures are depicted in Figure S1 of the Supplementary Information and their lattice parameters and atomic coordinates under ambient pressure are summarized in Table S2.

To uncover the structural transition of ZrH_2 under high pressure, we constructed the enthalpy-pressure curve of ZrH_2 without zero-point energy within the pressure range of 0–500 GPa, as displayed Fig. 1 (a). The most stable phase is the $I4/mmm$ phase under ambient pressure, whose enthalpy is approximately 9.77 meV/atom lower than that of the cubic $Fm-3m$ phase. This phenomenon can be attributed to the distortion from high-temperature $Fm-3m$ structure to low-temperature $I4/mmm$ phase, which can be also seen in previous investigations [26–30]. The

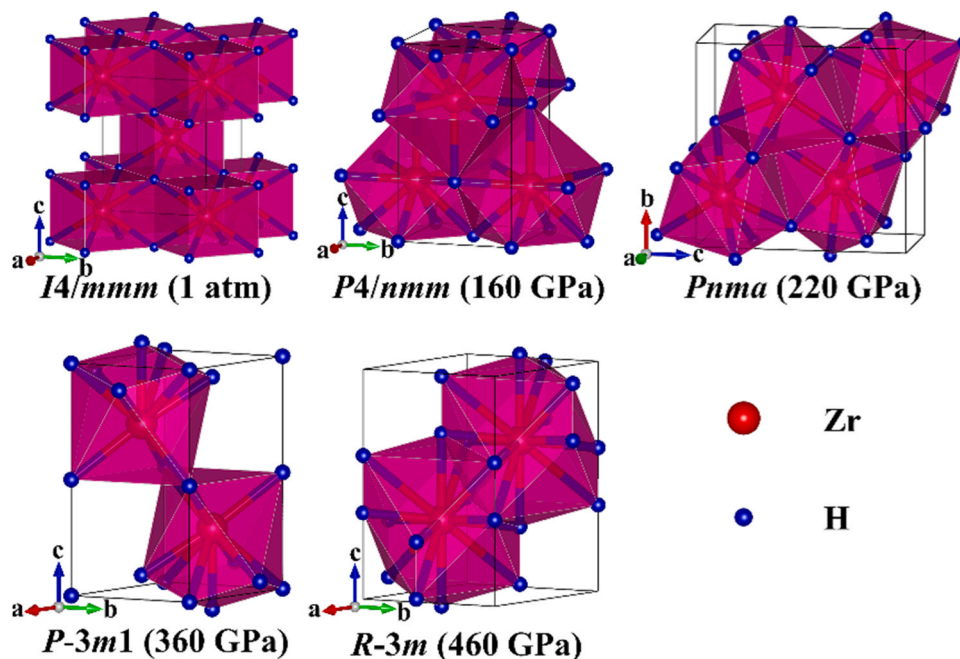


Fig. 2. Ground-state structures of ZrH_2 under different pressures.

$Pnma$ phase becomes stable from about 203 GPa, which is the same as the ground-state structures of TaH_2 [45] and VH_2 [8] under high pressure. Upon further compression, the $Pnma$ phase undergoes a structural transition into the $P-3m1$ phase at about 305 GPa. Furthermore, the $R-3m$ phase is found to be stable at 476 GPa. In short, the high-pressure transition sequence of ZrH_2 is $I4/mmm \rightarrow Pnma \rightarrow P-3m1 \rightarrow R-3m$ without considering zero-point energy effect with the transformation pressures of 203, 305 and 476 GPa, respectively. In the present work, the energetically stable phase is always the $I4/mmm$ configuration under 0–150 GPa, which is in agreement with previous theoretical research [31]. On the basis of the fact that the zero-point energy effect is enough to affect the relative stabilities of hydrides [46,47], we further probe into the structural transformation of ZrH_2 taking account into zero-point energy (ZPE) under the same pressure range (see Fig. 1(b)). With the zero-point energy effect, the transformation order converts into $I4/mmm \rightarrow P4/nmm \rightarrow Pnma \rightarrow P-3m1 \rightarrow R-3m$ and the transition pressures were determined as 117, 212, 231 and 444 GPa, respectively. Hence, the zero-point energy influences on the structural transition of ZrH_2 can be summarized into two aspects, namely, phase transition order and structural transformation pressure. On the one hand, the $P4/nmm$ structure is inserted between $I4/mmm$ and $Pnma$, which remains stable within the pressure range of 117–212 GPa. On the other hand, the transition pressure for the $Pnma$ phase become higher while those for the $P-3m1$ and $R-3m$ structures shrink. Additionally, the relative stability of ZrH_2 in comparison with other Zr-H compounds were constructed by calculating the formation enthalpies (ΔH_f) of known Zr-H compound without and with ZPE at 1 atm, 160 GPa, 220 GPa, 360 GPa, 460 GPa and 500 GPa, as displayed in Figure S2. Under atmospheric pressure, the calculated ΔH_f of two Zr-H compounds (ZrH and $I4/mmm\text{-ZrH}_2$) are precisely located at the curves of convex hull, which is in excellent agreement with the previous results proposed by Li et al. [31], indicating that the $I4/mmm\text{-ZrH}_2$ thermodynamic stability at ambient pressure. At the same time, from Figure S2(a), the ΔH_f of $I4/mmm\text{-ZrH}_2$ at 160 GPa, $Pnma\text{-ZrH}_2$ at 220 GPa, $P-3m1\text{-ZrH}_2$ at 360 GPa, $P-3m1\text{-ZrH}_2$ at 460 GPa and $R-3m\text{-ZrH}_2$ at 500 GPa without ZPE can reach -0.931 , -0.855 , -0.708 , -0.605 and -0.571 eV/atom, respectively. Moreover, from Figure S2(b), the ΔH_f of $P4/nmm\text{-ZrH}_2$ at 160 GPa, $Pnma\text{-ZrH}_2$ at 220 GPa, $P-3m1\text{-ZrH}_2$ at 360 GPa, $R-3m\text{-ZrH}_2$ at 460 GPa and $R-3m\text{-ZrH}_2$ at 500 GPa with ZPE can reach -0.904 , -0.823 , -0.668 , -0.560 and

-0.525 eV/atom, respectively. These two consequences above indicate that all the ZrH_2 phases without and with ZPE possesses lower negative formation enthalpies under the corresponding pressures of 160, 220, 360, 460 and 500 GPa, slightly above the convex hull, verifying their excellent thermodynamic stability [48]. As these five structures ($I4/mmm$, $P4/nmm$, $Pnma$, $P-3m1$ and $R-3m$) are considered as the ground-state phases under the corresponding pressures, we focus on their structural and related properties.

The crystal structures of $I4/mmm$ at 1 atm, $P4/nmm$ at 160 GPa, $Pnma$ at 220 GPa, $P-3m1$ at 360 GPa and $R-3m$ at 460 GPa for ZrH_2 are listed in Fig. 2 and their structural parameters are summarized in Table S3. The lattice parameters ($a = b = 3.54$ Å and $c = 4.41$ Å) of the $I4/mmm$ phase at 1 atm are consistent with those in experiment ($a = b = 3.52$ Å, $c = 4.45$ Å) [20] and in theory ($a = b = 3.54$ Å, $c = 4.41$ Å) [49]. Obviously, the H coordination numbers around Zr atoms for $I4/mmm$ at 1 atm, $P4/nmm$ at 160 GPa, $Pnma$ at 220 GPa, $P-3m1$ at 360 GPa and $R-3m$ at 460 GPa are 8, 9, 9, 10 and 14, respectively, suggesting the augmentation for the H coordination numbers of Zr atoms with the increasing pressure. This high-pressure effect can also be observed in some dihydrides and dioxides [50–52]. This fourteen-fold coordinated phase in $R-3m\text{-ZrH}_2$ is firstly found in transition-metal dihydrides. The nearest H–H distances of $I4/mmm$ at 1 atm, $P4/nmm$ at 160 GPa, $Pnma$ at 220 GPa, $P-3m1$ at 360 GPa and $R-3m$ at 460 GPa are 2.20, 1.91, 1.82, 1.49 and 1.21 Å, respectively. Simultaneously, the smallest Zr–H separations for these five phases above are 2.08, 1.80, 1.79, 1.70 and 1.68 Å, respectively. These two phenomena that the Zr–H and H–H separations become decrease with the increasing pressure is the main effect for the increasing pressure. Obviously, the nearest H–H distances in these five stable structures is 1.21 Å, which is longer than that of free H_2 molecule (0.75 Å) or H_3 unit (< 1.00 Å). The result above suggests that no covalent H_2 or H_3 units form in these five structures [53–56].

3.2. Electronic Properties

The electronic band structure in conjunction with total and partial density of states for five considered configurations ($I4/mmm$ at 1 atm, $P4/nmm$ at 160 GPa, $Pnma$ at 220 GPa, $P-3m1$ at 360 GPa and $R-3m$ at 460 GPa) were roundly computed to uncover their electronic properties, as drawn in Figure S3. The evident overlap between the conduction and

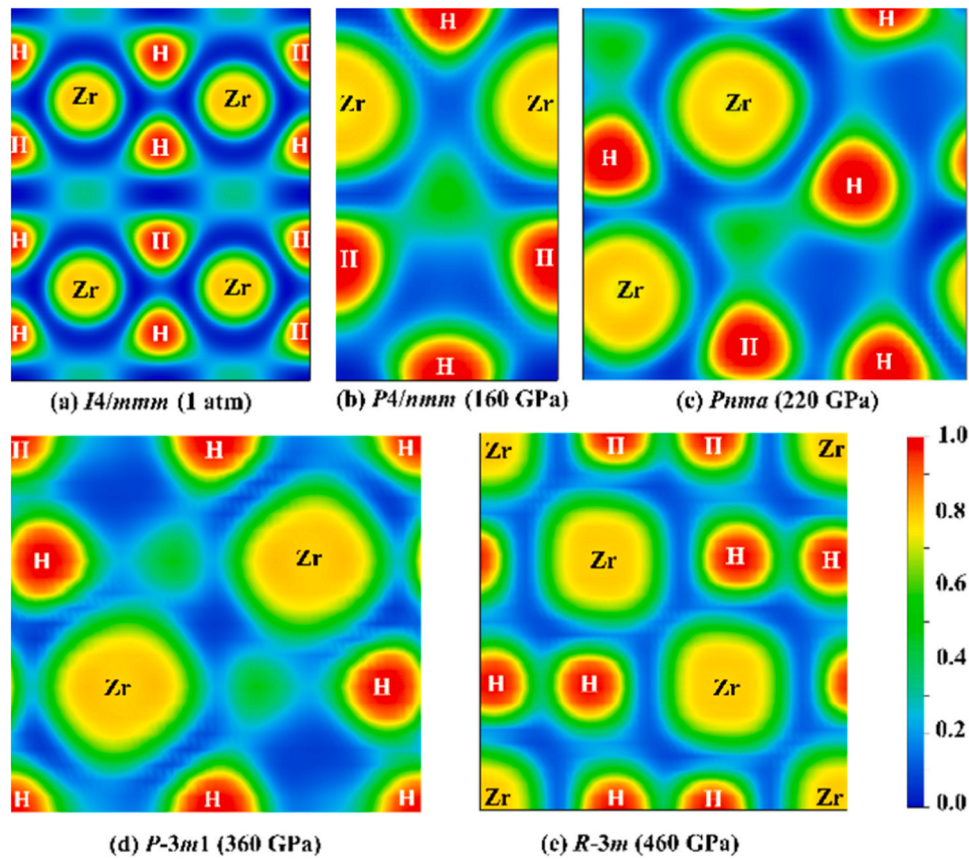


Fig. 3. Electron localization function of ZrH_2 under different pressures.

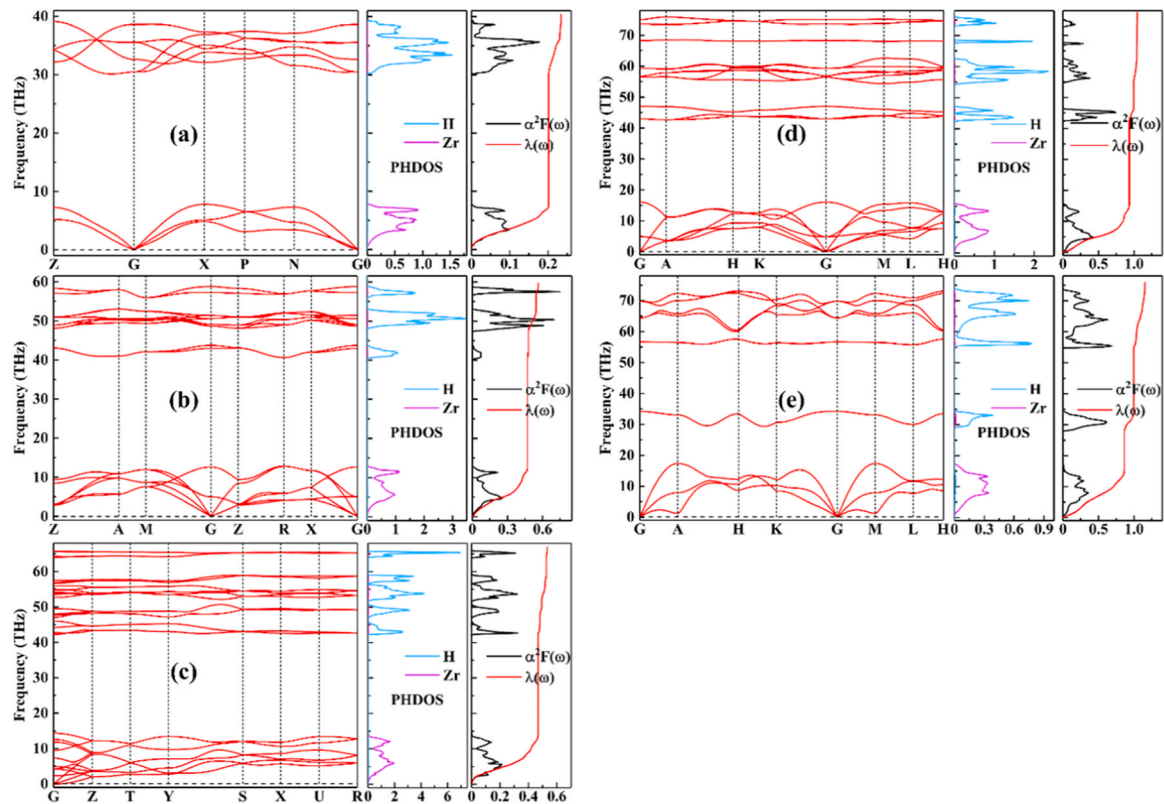


Fig. 4. Phonon dispersion curves, atom-projected phonon density of states, Eliashberg spectral function $\alpha^2F(\omega)$ and the electron-phonon coupling parameter λ of ZrH_2 for (a) $I4/mmm$ at 1 atm (b) $P4/nmm$ at 160 GPa (c) $Pnma$ at 220 GPa (d) $P-3m1$ at 360 GPa and (e) $R-3m$ at 460 GPa.

Table 1

The calculated, the logarithmic average phonon frequency $\omega_{\log}$, electron-phonon coupling parameters λ and superconducting critical temperatures T_c of ZrH₂.

Phase	P (GPa)	$\omega_{\log}$ (K)	λ	T_c (K)		
				$\mu^* = 0.10$	$\mu^* = 0.13$	Exp. ^a
<i>I4/mmm</i>	1 atm	216.40	0.207	0.00	0.00	< 1.00
	40	337.23	0.166	0.00	0.00	
	80	374.85	0.154	0.00	0.00	
	100	400.67	0.162	0.00	0.00	
<i>P4/nmm</i>	120	287.57	0.506	3.66	2.24	
	140	261.95	0.550	4.52	2.98	
	160	278.13	0.593	6.15	4.30	
	180	224.74	0.686	7.43	5.64	
	200	189.43	0.813	9.17	7.45	
	220	314.07	0.601	7.22	5.08	
<i>Pnma</i>	260	168.10	1.261	15.97	14.26	
	300	197.49	1.229	18.21	16.21	
	360	260.67	1.034	19.08	16.49	
	400	206.57	1.137	17.26	15.19	
	440	242.36	1.132	20.14	17.70	
	460	261.43	1.250	24.59	21.94	
<i>R-3m</i>	500	330.81	1.214	30.03	26.68	

^a Reference [61].

the valence bands for these five structures reflects their metallic features. This phenomenon could also be substantiated by the adjacent density of states, where there exists large total density of state around the Fermi level. Additionally, the total density of states (TDOS) at the Fermi level is mainly contributed by the Zr-4d states, which indicates that the Zr-4d states play a significant role in the electronic behavior of this material near the Fermi level. For *I4/mmm* structure, the value of its TDOS at the Fermi level does not exceed 1 state/eV/f.u., which is similar to TiH₂ and HfH₂ at ambient pressure [19,21,22]. Under high pressure, the TDOS of the stable structures becomes larger, similar to VH₂, NbH₂, and HfH₂ [18,21,22]. From the projected density of states (PDOS), it is evident that the occupation of the five ground state structures are similar. Below the pseudogap, there is a dominant hybridization feature between the H-1s orbital and the Zr-4d orbitals, while the Zr-4d states dominate in this energy range around the Fermi level. This observation is consistent with the DOS features of VH₂, NbH₂, TiH₂ and HfH₂ [18,19,21,22].

To further probe into the bonding characters for these five structures (*I4/mmm* at 1 atm, *P4/nmm* at 160 GPa, *Pnma* at 220 GPa, *P-3m1* at 360 GPa and *R-3m* at 460 GPa), we calculated the electron localization function (ELF) [57], as presented in Fig. 3 through VESTA [58,59]. Among these five stable phases, there are almost no electrons localized between Zr and H atoms, indicating the ionic character of the Zr-H bonds. Furthermore, the Zr-H ionic nature were also confirmed with the Bader charge analysis (see Table S4). As is evident, there exists large amount of charge transferring from Zr to H atoms in five stable structures, which could be attributed to the discrepancy of element electronegativity. According to the periodic law, the electronegativity value of the H element (2.2) is extremely larger than that of the Zr element (1.33), illustrating that the H atom should receive charge from the Zr atom. From Fig. 3, the ELF values at the center of the H-H connection in *I4/mmm* at 1 atm, *P4/nmm* at 160 GPa and *Pnma* at 220 GPa, *P-3m1* at 360 GPa and *R-3m* at 460 GPa are no more than 0.5, declaring weak interacting for neighboring H-H.

To further elucidate the bonding nature of the H-H connection in these five focused configurations (*I4/mmm* at 1 atm, *P4/nmm* at 160 GPa, *Pnma* at 220 GPa, *P-3m1* at 360 GPa and *R-3m* at 460 GPa), we calculated their Crystal Orbital Hamilton Population and the integrated Crystal Orbital Hamilton Population (-ICOHP) of H-H, as summarized in Table S5. Apparently, the resulting -ICOHPs of H-H in *I4/mmm*, *P4/nmm* and *Pnma* are not exceeding 0.04 while the -ICOHPs for H-H in *P-3m1* and *R-3m* can reach 0.13 and 0.74. These consequences above indicate that the H-H interaction in *I4/mmm*, *P4/nmm* and *Pnma* is

weak whereas the H-H interaction should be relatively strong in the *P-3m1* and *R-3m* phases. In short, based on the analysis of bond lengths, ELF and -COHP, ZrH₂ can be classified as an ionic crystal due to the presence of zirconium-hydrogen ionic bonds.

3.3. Superconductivity

To further disclose the dynamical stability, the phonon dispersion curves and projected phonon density of states were calculated for *I4/mmm* at 1 atm, *P4/nmm* at 160 GPa, *Pnma* at 220 GPa, *P-3m1* at 360 GPa and *R-3m* at 460 GPa, as depicted in Fig. 4. Obviously, there are no imaginary phonon frequencies in the whole Brillouin zone, demonstrating that all the five phases are dynamically stable [60]. As expected, the frequency region can be divided into two separate regions. The high-frequency region is related to the vibration of H atom while the low-frequency band corresponds to the vibration of Zr atom, which can be ascribed into the extremely lighter atom mass of hydrogen atom than zirconium atom.

In general, EPC stands for electron-phonon coupling, which refers to the interaction between electrons and lattice vibrations (phonons) in a material. We also calculated the logarithmic average phonon frequency $\omega_{\log}$ and the electron-phonon coupling parameter λ for *I4/mmm*, *P4/nmm*, *Pnma*, *P-3m1* and *R-3m*, as tabulated in Table 1. The superconducting temperature T_c was predicted according to the Allen-Dynes

modified McMillan equation: $T_c = \frac{\omega_{\log}}{1.2} \exp \left[\frac{1.04(1+\lambda)}{\lambda - \mu^*(1-0.62\lambda)} \right]$ with the

Coulomb parameter set to $\mu^* = 0.1$ and 0.13. It is noteworthy that the superconductivity for *I4/mmm* phase at 0–100 GPa is profoundly absent because of the weak electron-phonon coupling. This conclusion agrees well with the experimental T_c result of ZrH₂ ($T_c < 1$ K) [61]. Furthermore, the T_c s for the *P4/nmm* phase at 200 GPa, the *Pnma* phase at 220 GPa, the *P-3m1* phase at 440 GPa and the *R-3m* phase at 500 GPa are 7.45–9.17 K, 5.08–7.22 K, 17.70–20.14 K and 26.68–30.03 K, respectively. It can be easily observed from Table 1 that the increasing pressure results in increasing in the electron-phonon coupling parameter λ . It is apparently seen that the calculated electron-phonon coupling parameters λ for the *P-3m1* and *R-3m* structures under the pressure ranges of 260–440 GPa and 460–500 GPa are higher than 1, indicating strong electron-phonon coupling. Therefore, compared with other ZrH₂ structures (*I4/mmm*, *P4/nmm* and *Pnma*), the predicted T_c values for the *P-3m1* and *R-3m* phases is significantly enhanced. Therefore, the estimated T_c values are strongly related with λ parameter, which can be attributed to strong electron-phonon coupling.

4. Conclusions

In summary, on the basis of structure prediction simulations with the USPEX method, various low-energy crystalline structures of zirconium dihydride are derived over the pressure range of 0–500 GPa. Considering the ZPE effects, the structural transition sequence of ZrH₂ is *I4/mmm* → *P4/nmm* → *Pnma* → *P-3m1* → *R-3m* with the structural transformation pressures of 117, 212, 231 and 444 GPa, respectively. Among the five ZrH₂ phases, *Pnma*, *P-3m1* and *R-3m* are disclosed for the first time. According to the analyze of the bond length, ELF and -COHP, these five stable configurations mainly comprise the Zr-H ionic bonding. Interestingly, the predicted superconducting critical temperature T_c of *Pnma* at 220 GPa, *P-3m1* at 440 GPa and *R-3m* at 500 GPa can reach 5.08–7.22 K, 17.70–20.14 K and 26.68–30.03 K, respectively. The findings provide detailed insights into the pressure-induced novel structures and bonding properties of ZrH₂, and also demonstrate the importance of strong electron-phonon coupling for the superconductivity of ZrH₂.

CRedit authorship contribution statement

Jinqun Zhang: Writing – review & editing, Supervision, Software,

Formal analysis. **Libiao Tang**: Writing – original draft, Visualization, Software, Formal analysis. **Panlong Kong**: Writing – review & editing, Supervision, Software, Formal analysis. **Yuanyuan Jin**: Writing – review & editing, Funding acquisition, Conceptualization. **Dongming Huo**: Funding acquisition, Formal analysis. **Song Li**: Supervision, Conceptualization. **Chuanzhao Zhang**: Writing – review & editing, Software, Funding acquisition, Conceptualization. **Fanguang Kuang**: Supervision, Funding acquisition, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.mtcomm.2024.109516](https://doi.org/10.1016/j.mtcomm.2024.109516).

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